

JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kinematic Equations & Free Fall	63 PYQs	Kinematics
Very High	Velocity-Time & Position-Time Graphs	58 PYQs	Graphical Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kinematic Equations & Free Fall	$v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s_n = u + \frac{a}{2}(2n - 1)$
Velocity-Time & Position-Time Graphs	$v = \frac{dx}{dt}$, $a = \frac{dv}{dt} = \frac{dv}{dx}$, $\text{Displacement} = \int v dt$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kinematic Equations & Free Fall (Very High Yield — 63 PYQs)

Core Principle: Uniformly accelerated rectilinear motion, vertical motion under gravity, and displacement in the n th second.

■ **Caution / JEE Pitfall:** Assign a consistent sign convention (e.g. upward positive, downward negative) at the beginning of problem solving.

■ **Speed Trick:** Stopping distance $d \propto u^2$. For an object dropped from rest, distance ratio in successive seconds is 1 : 3 : 5 : 7 (Galileo's odd numbers).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A gas balloon is going up with a constant velocity of 10m/s . When this balloon reached a height of 75 m , a stone is dropped from it and ..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begi..." [Ans: C]
- [2026 Evening Shift] Q#3 (MCQ): "Two cars A and B are moving in the same direction along a straight line with speeds 100km/h and 80km/h , respectively such that car ..." [Ans: C]

3.2 Velocity-Time & Position-Time Graphs (Very High Yield — 58 PYQs)

Core Principle: Interpretation of motion curves: slope of x - t is velocity, slope of v - t is acceleration, area under v - t is displacement.

■ **Caution / JEE Pitfall:** Area under speed-time curve gives distance (always positive), while area under velocity-time curve gives displacement.

■ **Speed Trick:** When a is given as a function of x , use $a = v(dv/dx)$ rather than (dv/dt) .

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s . The total distance travelled b..." [Ans: C]
- **[2026 Evening Shift] Q#2 (MCQ):** "The velocity (v) - Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of th..." [Ans: D]
- **[2026 Evening Shift] Q#4 (MCQ):** "A particle starts moving from time $t=0$ and its coordinate is given as $x(t)=4t^3-3t$ A. The particle returns to its original position (o..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A gas balloon is going up with a constant velocity of 10m/s . When this balloon reached a height of 75 m , a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is ____ m. (Take $g=10\text{m/s}^2$)

- (A) 85
- (B) 150
- (C) 129
- (D) 125 [Correct]

Correct Answer: D

Solution Analysis: When an object is dropped from a moving carrier (like the balloon), it gets the velocity of that carrier at the exact moment of release. Initial velocity of the stone (balloon) is $u(s)=10\text{m/s}$ (upwards). Initial height of the stone is $h=75\text{m}$. The acceleration due to gravity is $g=-10\text{m/s}^2$ (downwards). Using the second equation of motion for the stone's displacement along vertical direction, $y=u(y)t+(1/2)a(y)t^2$ Since the stone eventually hits the ground (at height 0), its displacement from the starting point (75 m) is $y= -75\text{ m}$. $-75=(10)t+(1/2)(-10)t^2$ Solving for t , we get $t=5\text{ s}$. In this time, the balloon has moved up with a constant velocity of 10m/s. So, the height of the balloon when the stone hits the ground is $75\text{ m} + (10\text{m/s}) \times (5\text{ s}) = 125\text{ m}$.

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#2)

The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s . The total distance travelled by the particle and the average velocity during this period are, respectively ____ .

- (A) 25 m and zero
- (B) 50 m and zero
- (C) 100 m and zero **[Correct]**
- (D) 100 m and 2.5m/s

Correct Answer: C

Solution Analysis: The displacement is the algebraic sum of the areas (assuming area above the t -axis as positive and area below as negative). And the distance is the sum of the magnitudes of the areas (assuming all areas as positive because distance can't be negative). The graph consists of two triangles: For triangle 1 from $t=0$ to $t=20$ s : Base $b_1=20$ s , Height $h_1=+5$ m/s So, area of triangle 1 is, $\text{Area}_1 = \frac{1}{2} \times 20 \times 5 = 50$ m For triangle 2 from $t=20$ to $t=40$ s : Base $b_2=40-20=20$ s , Height $h_2=-5$ m/s So, area of triangle 2 is, $\text{Area}_2 = \frac{1}{2} \times 20 \times (-5) = -50$ m